

Sahithi Kalva

Senior Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with 6+ years of experience designing, building, and optimizing **scalable data platforms** and **pipelines** across **cloud** and **distributed environments**. Proven expertise in **end-to-end ETL/ELT architecture**, **batch** and **real-time data processing**, and **modern data stack implementations** using **AWS, Azure, Snowflake, dbt, and Databricks**. Strong background in **Big Data technologies** including **Apache Spark, Kafka, Airflow, Flink, and Hadoop**, supporting **petabyte-scale analytics** and **AI/ML workloads**. Demonstrated success in **advanced SQL performance tuning**, **Python-based data engineering frameworks**, **data modeling**, **data quality automation**, and **CI/CD pipeline implementation**. Adept at partnering with cross-functional teams to deliver production-grade data solutions that power **analytics, machine learning, and enterprise decision-making**.

TECHNICAL SKILLS

Programming & Scripting: Python, SQL, Scala, Shell Scripting (Unix/Linux)

Big Data & Distributed Computing: Apache Spark, PySpark, Apache Hadoop, Apache Hive, Apache Sqoop, Amazon EMR

Data Engineering & ETL Tools: dbt (Core), Apache Airflow, AWS Glue, Azure Data Factory, SSIS, ETL/ELT Pipelines

Cloud Platforms & Services: AWS (S3, EC2, Glue, Redshift, RDS), Azure (Databricks, Synapse Analytics, Data Lake, Azure ML)

Data Warehousing & Modeling: Snowflake, PostgreSQL, SQL Server, MySQL, Oracle, Data Vault 2.0, Dimensional Modeling

Streaming & Real-Time Data: Apache Kafka, Apache Flink, Spark Streaming, Amazon Kinesis, Azure Event Hubs

Machine Learning & Feature Engineering: PySpark ML, scikit-learn, TensorFlow, PyTorch, SageMaker, Feature Engineering

DevOps & DataOps: Git, GitHub, Jenkins, Docker, Kubernetes, Terraform, Azure DevOps, CI/CD Pipelines

Data Quality & APIs: Great Expectations, FastAPI, REST APIs, Data Validation, HIPAA Compliance

Methodologies & BI: Agile, Scrum, Software Development Life Cycle (SDLC), Power BI, Tableau

PROFESSIONAL EXPERIENCE

CARIAN | NJ | Senior Data Engineer

July 2025 – Present

- Architected and deployed **scalable cloud data platforms** using **AWS (S3, Glue, Redshift)** and the **modern data stack (Snowflake, dbt)**, integrating multiple structured and semi-structured data sources and reducing analytics delivery by **25%**.
- Designed and optimized **ETL/ELT pipelines** using **PySpark, SQL**, and **incremental data processing**, improving data freshness while reducing cloud compute costs for enterprise clients.
- Built **feature engineering pipelines** leveraging **dbt, SQL-based transformations**, and reproducible feature stores, supporting ML model training and improving predictive accuracy by **12%**.
- Developed **real-time data ingestion** and **processing pipelines** using **Spark Streaming**, and created **Tableau dashboards** to monitor live business KPIs and model performance metrics.
- Implemented end-to-end **CI/CD pipelines** for data and ML workflows using **Apache Airflow, Jenkins, GitLab CI/CD**, enabling automated testing, deployment, and monitoring of production data pipelines and reducing release errors by **15%**.

Meta | CA | Data Engineer

August 2024 – July 2025

- Developed **high-throughput, distributed data pipelines** using **Python** and internal Meta frameworks, orchestrated with **Apache Airflow**, improving large-scale data processing efficiency by **20%** across petabyte-scale systems.
- Optimized **real-time streaming pipelines** using **Apache Kafka** and **Apache Flink**, reducing data latency by **18%** and enabling near-real-time feature updates for production ML models.
- Built **scalable feature engineering pipelines** using **PySpark, dbt Core**, and Meta's AI infrastructure to process **billions of records**, increasing ML model training efficiency by **30%**.
- Designed and implemented **Spark-based ingestion and transformation systems** using **Scribe**, improving fault tolerance and reducing end-to-end processing time by **15%**.
- Automated build and release processes using **CI/CD pipelines (Jenkins, GitHub, Shell scripting)**, accelerating deployment cycles by **35%** while maintaining production stability.
- Led **Agile cross-functional collaboration**, mentoring **2 junior engineers** on **dbt best practices**, feature stores, and scalable data design, improving sprint velocity by **15%**.

- Designed and implemented a **Data Vault 2.0 architecture** on **Snowflake** using **dbt**, integrating **6+ enterprise source systems** to create a governed, auditable data platform supporting **1.4M+ provider records**.
- Built **secure, automated data ingestion pipelines** using **Apache NiFi** and **Python**, de-identifying **PHI** data and ensuring full **HIPAA compliance** across the data lifecycle.
- Implemented a **data quality** and **validation framework** using **Great Expectations**, enforcing **30+ business rules** and improving data trust scores from **78% to 90%**.
- Developed **curated data marts** and **RESTful APIs** using **FastAPI**, enabling actuarial and network strategy teams to self-serve key provider metrics and reducing ad-hoc data requests by **40%**.
- Managed and optimized **Apache Airflow DAGs** to orchestrate daily production workflows, consistently meeting SLAs for downstream analytics and reporting.

- Built and deployed **20+ enterprise-grade ETL pipelines** using **Azure Data Factory**, **Databricks**, **Synapse Analytics**, improving throughput by **18%** and optimizing cloud infrastructure costs.
- Developed **distributed data processing solutions** using **Apache Spark**, **Apache Kafka**, and **Apache Airflow** to process high-volume datasets (POS, IoT, transactional data), reducing analytics latency by **15%**.
- Modernized analytics platforms using **Snowflake** and **Power BI**, delivering secure, modular data models with automated lineage and reducing report delivery timelines by **30%**.
- Optimized legacy **Hadoop ecosystems (HDFS, Hive, Sqoop)**, decreasing batch runtimes by **25%** and supporting large-scale cloud migration initiatives.
- Created reusable **Python-based ETL frameworks** using **SQLAlchemy**, **Pandas**, and validation utilities, improving developer productivity by **30%** and reducing manual data errors by **45%**.

CERTIFICATIONS

- Python for Business and AI Development – **IBM**
- Career Essentials in Data Analysis – **Microsoft & LinkedIn**

EDUCATION

Master's in Management Information Technology and Administration Services | St. Francis College, Brooklyn, NY, USA.

Bachelor's in Computer Science | St. Pious X Degree and PG College, Hyderabad, Telangana, India.